IAT New Test 22

Question paper, answer key and worked solutions

60 questions 180 minutes 240 marks

Biology 1–15; Chemistry 16–30; Mathematics 31–45; Physics 46–60.

Correct +4; incorrect -1; unattempted 0. All 60 questions are available.

This is a provisional 2026-pattern authoring draft for 2027 preparation. Independent academic review is pending.

Instructions

Each item has exactly one correct option. Use the separate solution section for review after attempting the paper. **Figures and equations are part of the questions.**

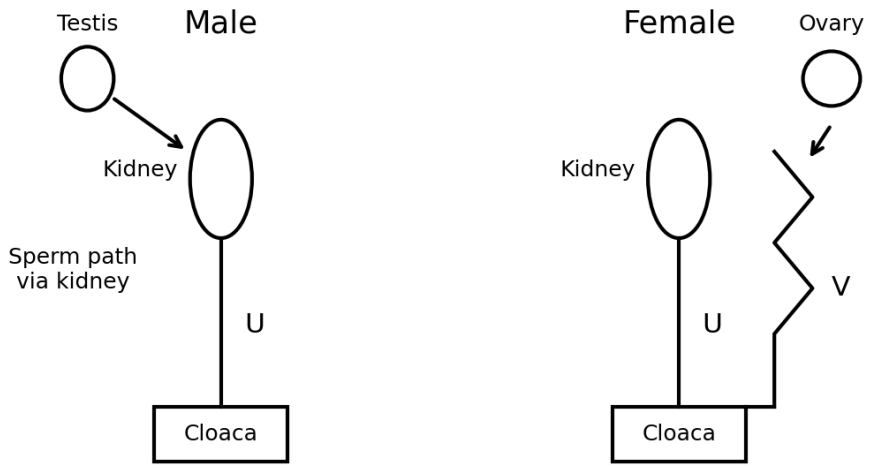
Biology

Q1 Three animals are examined. P has a water-canal system and cellular organisation without true tissues. Q has radial symmetry and two germ layers with tissue-level organisation. R is bilaterally symmetrical, triploblastic and has a pseudocoelom. Which assignment uses all these observations correctly?

A) P cnidarian; Q sponge; R annelid
B) P sponge; Q ctenophore; R flatworm
C) P sponge; Q cnidarian; R roundworm
D) P roundworm; Q cnidarian; R sponge

Q2 The adult-frog diagrams show male and female posterior duct systems. In the male, kidney-associated duct U receives the sperm-conducting pathway; in the female, V is an oviduct separate from U. Both panels end in a cloaca. Which interpretation is consistent with the diagram and normal frog reproduction?

Adult frog: posterior duct schematics



Separate female oviduct; shared terminal chamber. Not to scale.

A) Male U carries urine and sperm, female U carries urine, and a common cloaca does not imply internal fertilisation
B) U is an oviduct in both sexes, so the kidney releases eggs
C) Female V carries urine and sperm, while male U carries only digestive waste
D) The common cloaca proves that the frog normally fertilises eggs internally

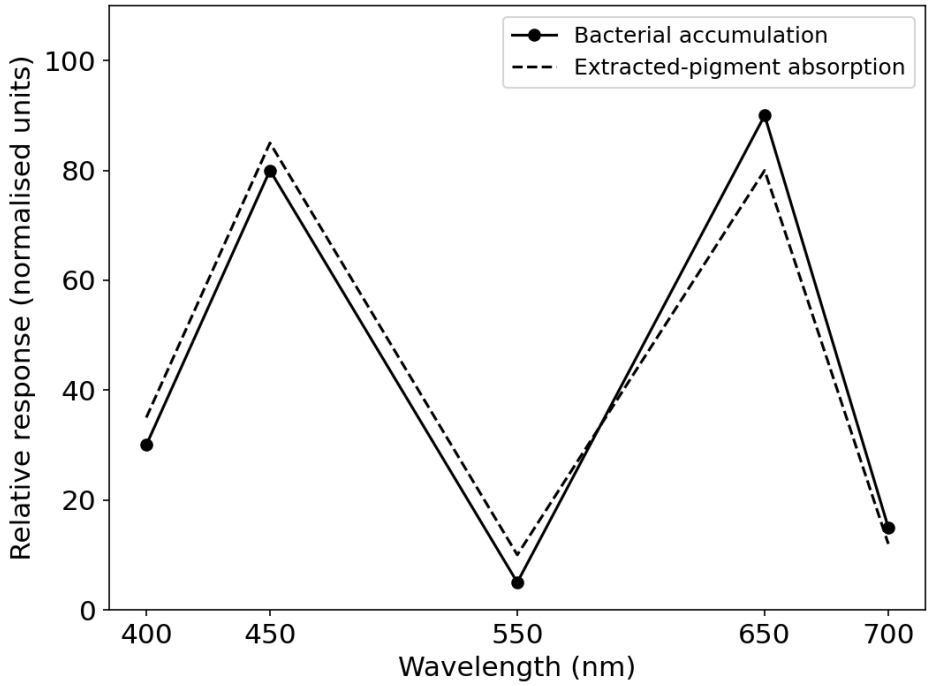
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Q3 A bacterial preparation contains a small circular DNA molecule separate from its main chromosome, non-membranous phosphate-storage granules, and 70S ribosomes. Which description distinguishes these structures without inventing eukaryotic organelles?

A) All three are membrane-bound organelles, because each has a cellular function
B) The DNA circle is a nucleus and the granules are lysosomes
C) The ribosomes are DNA-storage inclusions and the DNA circle is a mitochondrion
D) The DNA circle can be a plasmid, the granules are inclusions, and the ribosomes synthesise proteins

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Q4 An Engelmann-type preparation receives equal incident photon flux in the wavelength bands shown. Aerobic bacteria accumulate near the algal regions illuminated at 450 and 650 nm, but scarcely at 550 nm. Without the living alga they show no such band preference. An extracted-pigment absorption trace is also shown. Which conclusion is supported?



A) The bacteria demonstrate that green light can never be absorbed by any pigment
B) The oxygen-production action pattern broadly matches pigment absorption, without proving that one pigment alone accounts for all photosynthesis
C) The bacteria directly measure carbon-fixation enzymes absorbing blue light
D) The control proves that bacterial attraction is independent of oxygen production

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Q5 After a fall in renal perfusion, a subject activates renin release. Later, excessive atrial filling promotes atrial natriuretic factor release. In an ideal comparison ADH-dependent water permeability is held unchanged. Which paired response best explains the opposing effects on blood volume and pressure?
A) Renin directly inserts ADH water channels; atrial factor converts sodium into water
B) Both signals promote identical sodium retention, so their effects cannot oppose
C) Renin–angiotensin–aldosterone favours pressure and sodium retention; atrial factor opposes this retention and vasoconstriction
D) Atrial factor increases renin secretion to cause further pressure elevation

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Q6 A developing seed normally contains both endosperm and a persistent maternal nucellus called perisperm. In a limited experiment, egg fertilisation succeeds but fusion of a sperm with the polar nuclei is prevented; maternal nucellar tissue remains viable. Which tissue-origin prediction follows directly, without assuming that the abnormal seed will mature normally?
A) Normal endosperm initiation fails, while the retained nucellus can still be perisperm
B) Both storage tissues must have paternal DNA because both nourish the embryo
C) Perisperm must become endosperm when polar fusion fails
D) Successful egg fertilisation automatically produces both tissues from the zygote

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Q7 In a transformation experiment, purified material from heat-killed virulent bacteria transforms viable nonvirulent recipients after protease or RNase treatment, but not after DNase treatment. In a separate phage experiment, 32P-associated material enters bacteria predominantly, whereas most 35S remains outside. Which conclusion respects what these controls establish?
A) The two experiments prove that every virus and every organism must use DNA
B) DNase must have killed the recipient cells even though their viability control is normal
C) The isotope experiment by itself shows that all phosphorus-containing cell molecules are hereditary
D) The combined evidence supports DNA as the transforming and transferred genetic material in these systems, without excluding RNA genomes elsewhere

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Q8 In matched crop trials, inoculant P contains Trichoderma and reduces a soil-borne fungal disease without adding a nitrogen source. Inoculant Q contains a free-living nitrogen-fixing bacterium and increases available nitrogen without suppressing that pathogen. Both improve yield. Which interpretation distinguishes their demonstrated roles?
A) Both must be chemical fertilisers because final yield increased
B) P is a biofertiliser only, and Q acts by killing the fungus
C) P supplies biocontrol and Q biofertilisation; a common yield outcome does not establish a common mechanism
D) Q must form legume root nodules because every nitrogen fixer requires them

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Q9 A plant produces a double-stranded RNA matching a pest transcript T. Feeding pests show reduced T mRNA and protein, unchanged genomic T sequence, and reduced damage. A pest line expressing the same protein from a sufficiently sequence-divergent transcript T* retains that protein and survives feeding; controls confirm similar uptake of plant material. Which inference best combines these observations?
A) The treatment necessarily deletes the pest's T gene from every chromosome
B) Sequence-dependent RNA silencing can explain the effect; the T* bypass links suppression of this target protein to pest damage without establishing a permanent DNA edit
C) Double-stranded RNA must change the amino-acid sequence of every protein in the pest
D) The T* result proves that ingestion never occurred in the surviving pest line

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Q10 Equal litter masses are incubated with decomposers. P is lignin-rich and nitrogen-poor in cool dry conditions; Q is nitrogen-rich in warm moist aerated conditions; R is the same litter as Q at the same temperature and moisture but without oxygen. Which prediction and caution are best supported?
A) P must decompose fastest because lignin is an easily available sugar
B) R must exceed Q because removing oxygen always accelerates decomposition
C) Q must produce no mineral nutrients because decomposition only fragments litter
D) Q is expected to decompose faster than P; comparing Q with R isolates oxygen availability, whereas P versus Q confounds litter chemistry and climate

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Q11 A eukaryotic nuclear primary transcript contains both exons and introns. Before normal translation, a 5-prime cap and a 3-prime poly-A tail are added and the introns are removed. Which event directly joins the retained exon sequences?
A) Splicing
B) Charging a tRNA
C) Peptide-bond formation
D) DNA replication

Q12 A Human Genome Project-style reference sequence lists DNA bases and annotated candidate genes. Two individuals share a listed coding sequence but differ in phenotype. Which response correctly describes what the reference can and cannot establish?
A) The reference directly specifies every person's environment, so the difference must be an observation error
B) Every annotated gene is expressed equally in every tissue, so phenotype is fixed by the reference alone
C) The reference supports gene location and sequence comparisons, while expression, other variants and environment can still contribute to phenotype
D) The phenotype difference proves that the reference contains no useful genetic information

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Q13 At respiring tissues, CO_2 enters blood and much is converted to bicarbonate through carbonic-anhydrase-associated reactions. At the lungs the reverse conversion supplies CO_2 for exhalation. Which statement correctly links the two sites?
A) Most transported CO_2 must remain physically dissolved throughout because bicarbonate cannot be reconverted
B) Oxygen is converted directly into bicarbonate in tissues and back into oxygen in lungs
C) Carbonic anhydrase permanently removes CO_2 carbon from the transport system
D) Reversible conversion allows much CO_2 to travel as bicarbonate and be regenerated as CO_2 at the lungs

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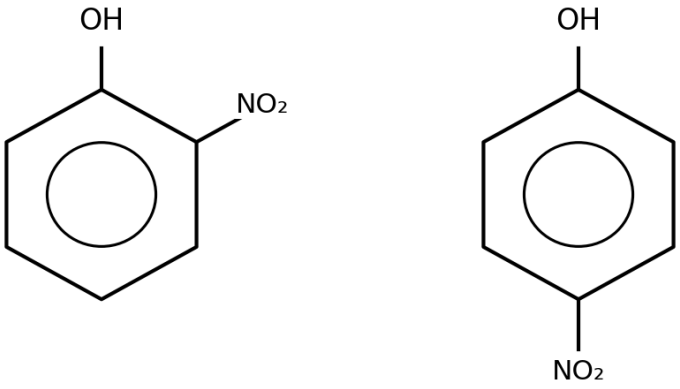
Q14 An enzyme loses activity in urea without peptide-chain cleavage. A reducing treatment also breaks its disulfide links. After both treatments, preparation X is allowed to refold before disulfides are re-formed and regains high activity. In Y, disulfides are re-formed while urea remains; after urea removal activity is low, but reducing and re-forming the links during refolding restores it. Chain sequencing is unchanged. Which explanation best fits every stage?
A) The same primary sequence can be trapped by non-native disulfide links; restoring appropriate folding and linkage can restore function
B) Disulfides are peptide bonds, so their re-formation necessarily makes a new primary sequence
C) Urea must remove the enzyme's gene, and oxidation repairs that gene
D) The original amino-acid sequence is irreversibly destroyed in Y

Q15 A preparation completely oxidises only glucose and a fatty acid of formula $C_{16}H_{32}O_2$, with no carbon retained in biomass. It consumes 29 mmol O_2 and releases 22 mmol CO_2. What fraction of the consumed oxygen is attributable to glucose oxidation? Treat both substrate amounts as unknown and use complete oxidation stoichiometry.
A) 22/29
B) 6/29
C) 1/2
D) 16/29

Chemistry

Q16 The structures P and Q are shown. Unknown samples X and Y correspond to these two isomers. An OH···O interaction in X persists after large dilution in an inert solvent; Y instead loses intermolecular association on dilution. Masking the OH donor by O-methylation removes those respective interactions. Both original vapours contain molecules of molar mass 139 g mol−1, and X is more readily steam-distilled. Which combined interpretation fits the geometry and controls?

Nitrophenol positional isomers



P: ortho Q: para

A) X is Q and Y is P; persistence on dilution proves an intermolecular network
B) X is P and Y is Q; equal vapour molar masses require equal liquid intermolecular forces
C) X is P and Y is Q; internal hydrogen bonding can persist on dilution while a less interconnected liquid is more volatile
D) X is P and Y is Q; masking OH proves that hydrogen bonding permanently changes the covalent molecular formula

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Q17 In a metal carbonyl, CO donates electron density from a filled orbital toward the metal, while the metal can return density from filled d orbitals into a CO antibonding orbital. Which joint bond trend is expected when the second interaction is strengthened within this model?
A) Metal–carbon bonding strengthens while the internal carbon–oxygen bond weakens
B) Both bonds necessarily weaken because every donation removes a bonding pair
C) The C–O bond strengthens because its antibonding occupancy rises
D) No interaction is possible because CO has no donor electron pair

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Q18 An insulated vessel has heat capacity 50 J K−1 and contains 10.0 g ice at 0 degrees C. It receives 4000 J. Use latent heat of fusion 334 J g−1 and liquid-water specific heat 4.20 J g−1 K−1, and neglect all other heat capacities and losses. Which final state follows?
A) Some ice remains at 0 degrees C
B) All ice melts; temperature is 4000/92 degrees C
C) All ice melts; temperature is 660/42 degrees C
D) All ice melts; temperature is 660/92 degrees C

Q19 A molecule contains a ketone and a separate cyclic acetal. The ketone must become a methylene group while the acetal remains protected. Assume the molecule tolerates heating and that only the stated functional groups react. Which choice and explanation best meet the requirement?
A) Zn(Hg)/concentrated HCl, because acetals require base for hydrolysis
B) Hydrazine followed by KOH and heat, because Wolff–Kishner reduction removes the ketone oxygen under basic conditions while the acetal resists base
C) NaBH4 followed by aqueous acid, because NaBH4 directly replaces C=O by CH2
D) Aqueous acid alone, because acetal hydrolysis also reduces the ketone

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Q20 Equal iron surfaces are coated with zinc or copper and then scratched to expose iron while coating and iron remain electrically connected in moist electrolyte. Standard reduction potentials are Zn2+/Zn=-0.76 V, Fe2+/Fe=-0.44 V and Cu2+/Cu=+0.34 V. Which comparison follows from galvanic action at the scratches?
A) Both coatings protect equally because neither scratch changes the coating metal’s identity
B) Copper sacrifices itself preferentially, whereas zinc makes iron the anode
C) Zinc can remain sacrificial to iron; copper can instead make exposed iron the anodic partner
D) Iron must remain cathodic against both metals because it is beneath a coating

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Q21 An amide CH3–13CO–NH2 has only its carbonyl carbon labelled 13C. Route P treats it with Br2/aqueous NaOH, isolates the resulting primary amine, then applies ordinary nitrous-acid deamination. Route Q instead reduces the amide with LiAlH4, isolates the primary amine, and applies the same deamination. No carbon scrambling occurs. Which final alcohols and label locations are expected?
A) P gives labelled ethanol; Q gives unlabelled methanol
B) P gives unlabelled methanol; Q gives ethanol labelled at its OH-bearing carbon
C) Both give ethanol labelled at its methyl carbon
D) Both give unlabelled methanol because every amide reduction removes its carbonyl carbon

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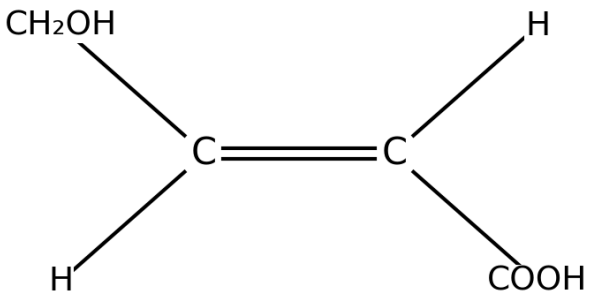
Q22 A 2.32 g sample of pure Fe3O4 is dissolved in acid without changing its iron oxidation states. The Fe2+ present is then titrated completely with 0.0200 M acidified KMnO4; Fe3+ does not react further. Use molar mass Fe3O4=232 g mol−1. What titrant volume is required, and how many moles of Fe3+ are present after titration?
A) 300 mL; 0.0300 mol
B) 100 mL; 0.0300 mol
C) 100 mL; 0.0100 mol
D) 20.0 mL; 0.0300 mol

Q23 At constant volume the reaction is 2A→3B. Over 30.0 s, [A] falls from 0.200 to 0.140 M, and no other reaction occurs. Using the stoichiometrically normalised average reaction rate, what are that rate and the increase in [B]?
A) 0.00200 M s−1; 0.0900 M
B) 0.00100 M s−1; 0.0600 M
C) 0.00200 M s−1; 0.0600 M
D) 0.00100 M s−1; 0.0900 M

Q24 For chlorobutanoic-acid isomers, the chlorine atom is placed at C2, C3 or C4, counting the carboxyl carbon as C1. Only the distance of this identical substituent from COOH changes. Which expected acidity order and conjugate-base order follow from the inductive effect?
A) Acid C2>C3>C4; conjugate-base basicity C2<C3<C4
B) Acid C2<C3<C4; conjugate-base basicity C2<C3<C4
C) All acidities identical because the molecular formula is unchanged
D) Acid C2>C3>C4; conjugate-base basicity C2>C3>C4

Q25 The structure in the figure is HOCH2–CH=CH–COOH, with the higher-priority groups CH2OH and COOH on opposite sides of the double bond. Which name respects the principal functional group, numbering and alkene configuration?

Unsaturated hydroxy acid: explicit substituents



A) (Z)-4-Hydroxybut-2-enoic acid
B) (E)-1-Hydroxybut-2-enoic acid
C) (E)-4-Hydroxybut-2-enoic acid
D) (E)-4-Carboxybut-2-en-1-ol

Q26 A sealed dilute solution contains 10.0 g of a molecular solute per litre and has osmotic pressure π=0.100 RT in consistent litre–mole units. Its membrane passes only solvent. The solute is volatile and contributes to the total equilibrium vapour pressure. Which inference is justified?
A) Osmotic molar mass is 100 g mol−1; the simple nonvolatile-solute vapour-pressure-lowering formula cannot be used on total pressure
B) Osmotic molar mass is negative because total vapour pressure can exceed the pure solvent pressure
C) Volatility alone makes the retained-solute osmotic measurement impossible
D) Both measurements must give 10.0 g mol−1 without accounting for solute vapour

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Q27 Complete combustion of 0.800 g of a compound containing only carbon and hydrogen gives 0.0500 mol CO2 and 0.100 mol H2O. The oxidant feed contains 0.150 mol total gas, 80.0 percent O2 and 20.0 percent inert gas; all carbon and hydrogen reach the stated products. What are the empirical formula and residual O2 amount? Use C=12,H=1.
A) CH2; 0.0450 mol
B) CH4; 0.0500 mol
C) CH4; 0.0200 mol
D) C2H4; 0.120 mol

Q28 Argon has atomic number 18 and relative atomic mass about 39.95; potassium has atomic number 19 and mass about 39.10. Which placement is consistent with the modern periodic law?
A) Potassium precedes argon because its relative atomic mass is lower
B) Argon precedes potassium because atomic number, not atomic mass, defines the modern ordering
C) They occupy one position because both masses are close to 40
D) Their order depends on how many neutrons a particular isotope contains

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Q29 A model compares prolonged impairment of fat absorption with a short period of low vitamin-C intake. Which statement best uses vitamin solubility without claiming that every water-soluble vitamin has identical storage behaviour?
A) All vitamins are stored equally, so solubility cannot affect deficiency risk
B) Vitamin C must require fat absorption because its name includes a letter
C) Fat malabsorption selectively prevents absorption of B vitamins but leaves A,D,E,K unaffected
D) Fat-soluble A,D,E,K are particularly vulnerable to impaired fat absorption; vitamin-C intake and limited reserves can produce a different time course

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Q30 A 0.100 M salt BH^+A^- is formed from weak base B with $K_b = 10^{-6}$ and weak acid HA with $K_a = 10^{-4}$. At 25 degrees C take $K_w = 10^{-14}$ and use the usual weak-acid/weak-base salt hydrolysis approximation. What pH and interpretation follow?
A) Approximately 6; BH^+ acidity exceeds the basic influence of A^-
B) Exactly 7; every 1:1 salt must be neutral
C) Approximately 8; the weaker parent base makes its conjugate acid weaker
D) Approximately 4; the salt behaves as 0.100 M strong acid

Mathematics

Q31 For the parabola $y^2=4x$, A=(1,2) and B=(9,6). What are the point at which the tangent is parallel to AB and the area of the parabolic segment bounded by AB and the parabola?
A) (4,4); 8/3
B) (4,4); 4/3
C) (2,2); 8/3
D) (4,4); 16/3

Q32 A differentiable f on (-1,∞) has f'(u)=(u-1)/(u+1). Let $g(x)=f(x^2)$ for all real x. Which classification of g's stationary points is correct?
A) Only x=0 is stationary, and it is a minimum
B) Only x=±1 are stationary, both maxima
C) There are two maxima and one minimum
D) There is a maximum at 0 and minima at -1 and 1

Q33 Evaluate $\int_0^1 \frac{x \ln(1+x)}{(1+x)^3}\,dx$.
A) (1-ln 2)/2
B) (5-6 ln 2)/16
C) (5-2 ln 2)/16
D) (3-2 ln 2)/16

Q34 How many relations on A={1,2,3} are symmetric and transitive but are not reflexive on A?
A) 3
B) 6
C) 10
D) 15

Q35 The equations x+2y+z=3, 2x+4y+2z=a and z=1 are considered over the real numbers. For which a is the system consistent, and how many solutions does it then have?
A) a=6; infinitely many
B) a=3; one
C) a=6; one
D) No a; the singular coefficient matrix is always inconsistent

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Q36 Independent Bernoulli trials are performed until the second failure. Success probabilities on trials 1,2,3 are 1/2,2/3,3/4; the fourth trial has a nonzero failure probability. Given that stopping occurs exactly on trial 4, what is the probability trial 1 was a success?
A) 2/3
B) 5/11
C) 1/2
D) 3/11

Q37 How many five-digit positive integers use each digit 0,1,2,3,4 exactly once and are divisible by 11?
A) 24
B) 12
C) 20
D) 16

Q38 For an integer n≥4, the coefficients of x^2 in (1+x)^n and (1+√5 x)^(n-3) are equal. What is the coefficient of x^3 in the first expansion minus that in the second?
A) 20-5√5
B) 20+5√5
C) 20-√5
D) 5√5-20

Q39 The sum of squares of the first n positive odd integers is two less than eight times their sum. What is the positive integer n?
A) 4
B) 5
C) 6
D) 8

Q40 Let $f(x)=x^2 \sin(1/x)$ for $x\neq 0$ and $f(0)=a$. Which statement about extension at zero is complete?
A) For a=0, f'(0)=0 and f' is continuous at zero
B) For a=0, f'(0)=0 but f' is not continuous there; no a makes the extension continuously differentiable at zero
C) No a makes f differentiable at zero
D) Every a makes f continuous, but none makes it differentiable

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Q41 Consider $y'=\sin y$ on the real line. Which are its constant solutions, and for the nonconstant solution with $y(0)=\pi/2$ and $0<y<\pi$, what is $y(\ln\sqrt{3})$?
A) Only y=0; value π/3
B) All y=nπ with integer n; value π/3
C) Only y=π; value 2π/3
D) All y=nπ with integer n; value 2π/3

Q42 From P=(1,2,3), a line parallel to (1,0,1) meets the plane x+y+z=0 at Q. What are Q and the distance PQ?
A) (-2,2,0); $3\sqrt{2}$
B) (-1,0,1); $2\sqrt{3}$
C) (0,2,2); $\sqrt{2}$
D) (-2,2,0); $2\sqrt{3}$

Q43 Let a=(1,1,0) and require $a\times v=(0,0,2)$. What is the minimum possible squared magnitude $|v|^2$ over all real solutions?
A) 1
B) 4
C) 2
D) 0

Q44 Grouped classes [0,2),[2,4),[4,6) have frequencies a,2,1. Use class midpoints and population variance. If a is a nonnegative integer and the variance is 56/25, what are a and the mean?
A) a=19/7; mean 12/5
B) a=2; mean 13/5
C) a=2; mean 3
D) a=1; mean 3

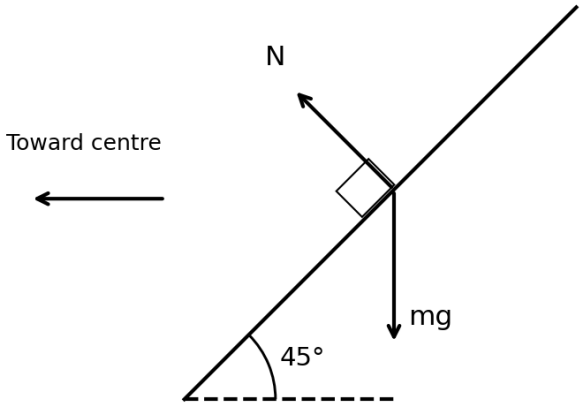
Q45 Using principal arcsin in [-π/2,π/2] and arccos in [0,π], consider all distinct real solutions of $\arcsin x + \arccos(2x^2-1) = 2\pi/3$. What is the sum of arcsin x over those solutions?
A) π/3
B) -π/9
C) 4π/9
D) 2π/9

Physics

Q46 Two measured masses are 20.600 g and 20.584 g. Their difference is divided by a measured volume 0.0250 cm3. Using the usual decimal-place rule for subtraction and significant-figure rule for division, what density should be reported without rounding the masses prematurely?
A) 0.6400 g cm−3, four significant figures
B) 0.6 g cm−3, one significant figure
C) 0.64 g cm−3, two significant figures
D) 0 g cm−3, because the rounded masses are equal

Q47 A road of radius 20 m is banked at 45 degrees. The coefficient of static friction is 1/3 and g=10 m s−2. What is the complete speed range for a vehicle to maintain the same horizontal circular path without slipping? Treat the car as a particle and allow static friction to act in either direction.

Banked circular road: transverse section



Static friction direction is to be determined.

A) 0≤v≤20 m s−1
B) 10≤v≤20 m s−1
C) 10≤v≤10√2 m s−1
D) v=10√2 m s−1 only

Q48 A uniform wire has original area 1.00 mm2, Young modulus 2.0×10^11 Pa and resistance 10.0 ohm. It is stretched elastically to longitudinal strain 0.010. In a first-order small-strain model its volume and resistivity are constant. What tensile force and resistance, to first order in strain, follow?
A) 2000 N; 10.2 ohm
B) 2000 N; 10.1 ohm
C) 200 N; 10.2 ohm
D) 2000 N; 9.8 ohm

Q49 A body at steady temperature T0 receives 10 W and loses 6 W by conduction, with the remaining loss by thermal radiation to a fixed environment. The conductive link is removed, incident power remains 10 W and radiation increases monotonically with body temperature. Which statement follows?
A) Radiation was 6 W, and the new steady temperature is lower
B) Radiation was 4 W, and the temperature must remain T0
C) Radiation was 10 W, and no temperature adjustment is needed
D) Radiation was 4 W, and the new steady temperature is higher than T0

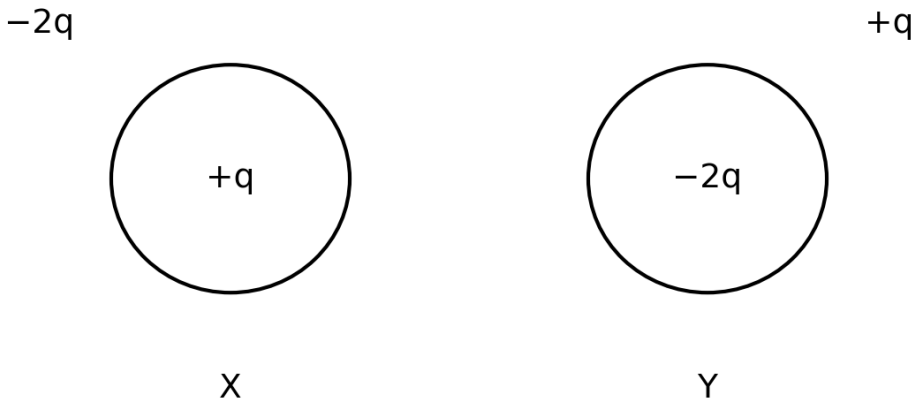
Q50 Two dilute packets of the same monatomic ideal gas have the same rest-frame pressure and number density. Packet A has zero mean laboratory velocity. Packet B has uniform laboratory drift speed U equal to its thermal rms speed w relative to its own mean motion. What is the ratio of mean molecular kinetic energy in the laboratory, B divided by A?
A) 1
B) 2
C) $\sqrt{2}$
D) 4

Q51 Two measured displacements are added. The first is $x1=A\cos(\omega t)$. At time $\tau=T/3$, where $T=2\pi/\omega$, a second independent oscillator begins contributing $x2=A\cos[\omega(t-\tau)]$. For $t\geq\tau$, what is the amplitude of $x1+x2$? The addition is a superposition of two signals, not an impulsive change to either oscillator.
A) 2A
B) $\sqrt{3}$ A
C) A
D) 0

Q52 A short upward pulse starts at the midpoint of an ideal string, moving right. The left end is fixed and the right end is free; speed is v and total length L. Neglect losses and pulse overlap away from an end. What are the signs of the first two returned pulses observed at the midpoint and their arrival times measured from launch?
A) First upward at L/v; second downward at 2L/v
B) First downward at L/v; second upward at 2L/v
C) Both downward, at L/(2v) and L/v
D) Both upward, at L/v and 2L/v

Q53 At stage X a closed Gaussian surface encloses +q while a separate -2q charge lies outside. At stage Y the +q charge is outside and -2q is inside; neither charge lies on the surface in either stage. The surface geometry is unchanged. What is final outward electric flux minus initial outward electric flux?

Same Gaussian surface geometry; changed charge positions

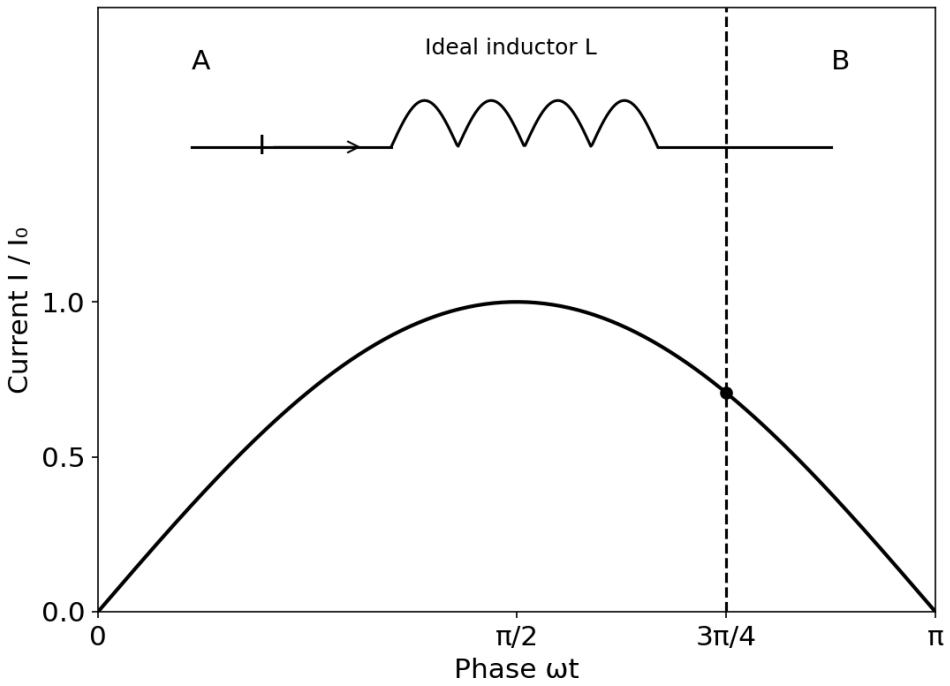


A) +3q/ε0
B) -q/ε0
C) 0
D) -3q/ε0

Q54 A potentiometer with unchanged uniform potential gradient balances two cells at 60.0 cm and 90.0 cm. Their internal resistances are different, but each balance is a true null with no current drawn from the tested cell. What is E1/E2, and why are their internal resistances irrelevant to that ratio?
A) 3/2; because resistance is always irrelevant in any voltage measurement
B) 2/3; at null the tested cell has no internal voltage drop
C) 1; because the same potentiometer wire is used
D) 2/3; because both tested cells carry identical nonzero current

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Q55 The ideal inductor in the diagram has current I=I0 sin(ωt) entering terminal A and leaving B. At t=3π/(4ω), which signs describe I, terminal voltage VA-VB and the rate of change of its stored energy?



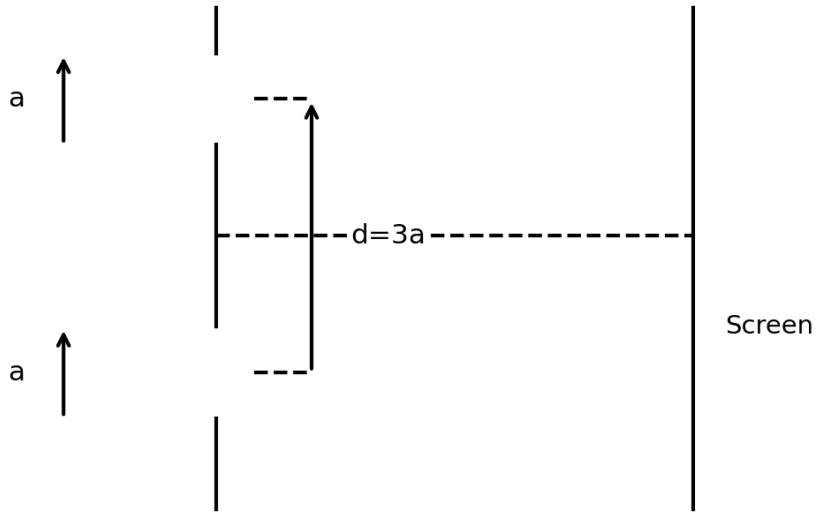
A) I positive; voltage positive; energy increasing
B) I negative; voltage negative; energy increasing
C) I positive; voltage negative; energy decreasing
D) I positive; voltage zero; energy constant

Q56 A transformer model uses the ideal induced-emf turns relation with Np/Ns=5 and primary voltage 100 V rms. Its secondary winding has resistance 1 ohm and supplies a 4 ohm resistor. Core loss is 20 W and all other losses are neglected. What are the delivered load power and efficiency?
A) 80 W; 80 percent
B) 64 W; 80 percent
C) 100 W; 64 percent
D) 64 W; 64 percent

Q57 An astronomical telescope has objective focal length 80 cm and eyepiece focal length 5 cm. It is initially normally adjusted for a distant object. The eyepiece alone is moved 1 cm toward the objective. What are the final image distance measured from the eyepiece with Cartesian signs and the magnitude of the angular magnification? Use paraxial angles and an eye close to the eyepiece.
A) -20 cm; 20
B) +20 cm; 20
C) -20 cm; 16
D) -4 cm; 20

Q58 Two identical slits of width a have centre separation d=3a. Use interference maxima d sinθ=mλ and single-slit diffraction minima a sinθ=nλ, with nonzero integer n. Which is the first missing noncentral interference order, and how many bright interference maxima remain strictly inside the central diffraction envelope?

Two finite slits: dimensions before the screen



A) Second order; three maxima
B) Third order; seven maxima
C) Third order; five maxima
D) Sixth order; five maxima

Q59 Two nonrelativistic particles have masses m and 4m and freely adjustable positive kinetic energies KA,KB with fixed sum K. Which division minimises the product of their de Broglie wavelengths, and what is λA/λB at that minimum?
A) KA=4K/5, KB=K/5; ratio 1
B) KA=KB=K/2; ratio 2
C) KA=KB=K/2; ratio 1
D) KA=K/5, KB=4K/5; ratio 4

Q60 A silicon crystal is doped with dilute pentavalent donors. In a uniform bulk region each donor is ionised once, no external charge has been added, and electron concentration greatly exceeds hole concentration. Which description preserves both carrier identity and bulk charge accounting?
A) The bulk must be negatively charged because mobile electrons are majority carriers
B) Positive donor ions move through the crystal as its main current carriers
C) The ionised donors are negatively charged and holes are the majority carriers
D) Electrons are majority mobile carriers; positive fixed donor ions balance their excess charge approximately in the neutral bulk

Answer key and worked solutions

Every solution explains the selected choice and the alternative errors.

Biology

Q1 / C Cellular organisation, germ layers and body-cavity type jointly classify three nonchordates

Answer: (C) Sponge, cnidarian and roundworm fit the respective character combinations.

Why: Poriferans have cellular-level organisation and a characteristic water-canal system. Cnidarians combine tissue-level organisation with radial symmetry and diploblasty. Roundworms are triploblastic bilateral animals with a pseudocoelom; this body cavity separates them from acoelomate flatworms and true-coelomate annelids. Classification therefore uses a combination of organisational and body-plan characters rather than symmetry alone.

Why not the others: (A) exchanges the canal-system animal with the true-tissue animal and gives R a true-coelomate identity. (B) A ctenophore can share some Q characters, but R cannot be an acoelomate flatworm, so the complete assignment fails. (D) gives the pseudocoelom to a sponge and ignores P's canal system.

Key point: A partial character match does not validate the complete classification.

Q2 / A Sex-specific frog ureter function separated from the meaning of a common cloaca

Answer: (A) Male and female ureter functions differ, and cloacal convergence does not establish internal fertilisation.

Why: In the male frog, the ureters function as urinogenital ducts, conducting urine and sperm toward the cloaca. In the female, the oviducts are separate from the ureters, so the urinary ducts do not acquire the same sperm-conducting role. The cloaca is a common terminal chamber for several systems; its presence describes anatomy, not the site where sperm and egg normally unite. Normal frog fertilisation is external in water.

Why not the others: (B) confuses renal ducts with oviducts and assigns egg production to kidneys. (C) gives the female oviduct urinary and male gamete functions absent from the stated arrangement. (D) infers a reproductive mechanism from a shared exit chamber, although the normal fertilisation site is external.

Key point: Follow each duct upstream before assigning its function.

Q3 / D Distinguish plasmid DNA, reserve inclusions and bacterial translation machinery

Answer: (D) These are a possible plasmid, storage inclusions and protein-synthesising ribosomes.

Why: A plasmid is an extrachromosomal DNA element rather than a membrane-enclosed nucleus. Prokaryotic inclusions can store reserve materials without a surrounding organelle membrane. Ribosomes perform translation and are also non-membranous; bacterial cytoplasmic ribosomes are 70S. Shared absence of an enclosing membrane does not make these three structures chemically or functionally equivalent.

Why not the others: (A) treats having a function as evidence of membrane enclosure. (B) assigns eukaryotic identities unsupported by either structure or function. (C) exchanges the information-storage and translation roles and invents a mitochondrion.

Key point: Identify a prokaryotic structure using its composition and function, not a eukaryotic name.

Q4 / B An oxygen-based spatial action pattern is compared with pigment absorption and an alga-free control

Answer: (B) The action pattern is consistent with pigment absorption, but it is not a single-pigment proof.

Why: Aerobic bacterial accumulation is an indirect indicator of oxygen-rich regions created by the illuminated alga. Peaks near blue and red bands agree broadly with strong absorption by photosynthetic pigments. The living-alga control helps exclude an intrinsic bacterial preference for those colours under the stated conditions. However, an extracted absorption trace and an oxygen-response pattern do not isolate every pigment's individual contribution in the intact cell.

Why not the others: (A) overgeneralises a low response in one band to every pigment and condition. (C) confuses the oxygen-based biological indicator with direct spectroscopy of carbon-fixation enzymes. (D) reverses the purpose of removing the oxygen-producing alga.

Key point: Distinguish an action spectrum from an absorption spectrum and interpret the control.

Q5 / C RAAS and atrial feedback are distinguished from an unchanged ADH pathway

Answer: (C) RAAS and atrial natriuretic signalling provide opposing regulatory influences.

Why: Reduced renal perfusion can trigger renin-dependent angiotensin formation. Angiotensin II contributes to vasoconstriction and aldosterone-associated sodium retention, which helps restore circulating volume and pressure. Increased atrial filling instead favours atrial natriuretic factor, which opposes the renin–angiotensin influence and favours sodium and water loss. Holding ADH-dependent permeability fixed isolates these sodium- and vascular-related effects rather than making every volume change an ADH response.

Why not the others: (A) confuses the renin cascade with ADH action and proposes a chemically impossible sodium conversion. (B) ignores the opposing feedback roles. (D) reverses the atrial signal's inhibitory influence on pressure-raising regulation.

Key point: Identify the initiating pressure signal and the regulated renal or vascular process separately.

Q6 / A A selective central-cell fertilisation block separates endosperm from maternal perisperm

Answer: (A) Endosperm and perisperm have different developmental origins.

Why: Typical angiosperm endosperm begins from the fertilisation product involving the central cell and the second sperm. Blocking that event prevents its normal initiation even when the egg has been fertilised separately. Perisperm is persistent nucellus, a maternal sporophytic tissue already present in the ovule; its identity does not depend on becoming a second fertilisation product. The prediction concerns tissue origin, not guaranteed survival of the entire abnormal seed.

Why not the others: (B) confuses nutritive function with parentage. (C) invents conversion between tissues with different origins. (D) assigns both tissues to the zygote and ignores double fertilisation's distinct products.

Key point: Shared storage function does not imply a shared fertilisation origin.

Q7 / D Selective degradation and isotope entry support DNA with explicit limits on generalisation

Answer: (D) DNA is supported in the tested systems, not established universally for every biological entity.

Why: Selective loss of transformation after DNase treatment identifies DNA as necessary for the activity of the purified transforming preparation when recipient viability is preserved. The phage tracer comparison independently distinguishes entry of phosphorus-labelled nucleic acid from most sulphur-labelled coat protein. Their agreement strengthens the DNA interpretation in these experimental systems. Neither experiment tested every virus, and RNA serves as genetic material in some viruses.

Why not the others: (A) extends experimental scope beyond the organisms tested. (B) contradicts the explicit viability control. (C) confuses a tracer's chemical distribution with a demonstration

that every labelled cellular molecule is hereditary.

Key point: Combine independent controls while retaining the limits of their experimental systems.

Q8 / C Matched microbial interventions separate pathogen suppression from nitrogen supply

Answer: (C) The measured mechanisms support biocontrol for P and biofertilisation for Q.

Why: Trichoderma is used as a biological control agent against plant pathogens. Here disease suppression, with no supplied nitrogen input, is the relevant observation. The second inoculant instead contributes nutrient availability through nitrogen fixation without the measured pathogen suppression. Free-living fixers need not form the specialised nodules associated with the Rhizobium–legume relationship. Yield integrates many processes and cannot identify which process caused an improvement on its own.

Why not the others: (A) equates an outcome with an inorganic chemical input. (B) reverses the measured disease and nutrient effects. (D) wrongly makes nodule formation universal among nitrogen fixers.

Key point: Diagnose a microbial application from its causal effect rather than yield alone.

Q9 / B Transcript-specific RNA interference with a sequence-divergent functional bypass

Answer: (B) A sequence-specific post-transcriptional mechanism and a functional bypass fit the evidence.

Step 1: The reduced matching mRNA and protein are consistent with RNA interference directed against T. The unchanged assayed genomic sequence gives no evidence for a DNA deletion.

Step 2: T* preserves the protein while changing the RNA sequence recognised by the silencing system. Its survival result therefore separates a sequence-targeting step from the function of the protein itself.

Step 3: Comparable feeding uptake weakens an explanation based solely on failure to ingest the material.

Why: The bypass supports a causal role for loss of the target protein, while the sequence comparison identifies the level at which targeting occurs. These observations do not require permanent genome modification.

Why not the others: (A) contradicts the genomic control. (C) confuses targeted transcript loss with universal protein recoding. (D) ignores the uptake measurement.

Key point: Separate target recognition, transcript abundance and downstream protein function.

Q10 / D Litter chemistry and environmental effects interpreted with a one-factor oxygen control

Answer: (D) Q combines favourable litter and environmental conditions, but only the Q–R comparison isolates oxygen.

Why: Decomposition generally proceeds faster with readily decomposable, nitrogen-rich litter and warm, adequately moist, aerated conditions; lignin-rich material and low temperature or dryness slow it. The P–Q contrast changes multiple factors, so it cannot quantify the effect of any one factor separately. Q and R instead hold litter, temperature and moisture fixed while changing oxygen, making them the appropriate pair for assessing that factor.

Why not the others: (A) misidentifies resistant lignin as an easily available sugar. (B) reverses the usual oxygen effect without evidence. (C) excludes mineralisation, an important part of decomposition.

Key point: Distinguish a multi-factor prediction from a controlled causal comparison.

Q11 / A Exon joining distinguished from end processing and translation

Answer: (A) Splicing joins the retained exons.

Why: The primary RNA transcript is processed before it serves as mature mRNA. Splicing removes intervening intron sequences and joins the exon portions in the RNA. Capping and polyadenylation modify the transcript's ends; they are separate processing operations and do not themselves join internal exons. Translation subsequently reads the mature mRNA to assemble a polypeptide, so it must not be substituted for the nuclear RNA-processing step.

Why not the others: (B) attaches an amino acid to tRNA rather than joining mRNA exons. (C) links amino acids during translation, not RNA segments. (D) copies DNA and does not perform RNA processing.

Key point: Keep transcript-end modification, internal splicing and translation distinct.

Q12 / C Reference-genome information separated from a complete individual phenotype prediction

Answer: (C) Sequence information is valuable without being a complete individual phenotype prediction.

Why: A reference genome provides an organised sequence framework for locating genes and comparing variants. It is not a record of every individual's regulatory state, expression pattern, other genomic differences or environmental history. Identical assayed coding sequence at one locus therefore does not guarantee identical complete biological context or phenotype. The observation limits an overextended claim about the reference; it does not negate the utility of sequencing.

Why not the others: (A) assigns environmental information to a DNA sequence resource. (B) ignores tissue-specific regulation and treats annotation as universal expression. (D) rejects valid sequence information because it cannot answer a broader question by itself.

Key point: A reference sequence enables comparisons but does not replace the rest of genotype-to-phenotype biology.

Q13 / D Tissue-to-lung reversal of bicarbonate-mediated carbon-dioxide transport

Answer: (D) Bicarbonate is a reversible transport form rather than permanent carbon disposal.

Why: Tissue CO2 can react with water to form carbonic acid and then bicarbonate and hydrogen ions. Carbonic anhydrase greatly speeds the relevant interconversion, especially in red cells. Pulmonary conditions favour the reverse direction, regenerating molecular CO2 that can diffuse into alveoli and leave with ventilation. Conversion during transport therefore does not prevent excretion of the same carbon at another site. It also does not mean that oxygen atoms supplied as O2 are the transported CO2 carbon source.

Why not the others: (A) excludes the major bicarbonate transport route. (B) confuses oxygen transport with carbon-dioxide chemistry. (C) treats reversible catalysis as irreversible destruction.

Key point: Follow the transported carbon through reversible chemical forms.

Q14 / A Order-dependent refolding controls distinguish primary sequence from a disulfide-linked structural trap

Answer: (A) The order of folding and disulfide formation can produce different functional structures without changing the peptide sequence.

Step 1: Unchanged sequencing and absence of peptide cleavage separate primary-sequence integrity from loss of activity. Urea can disrupt higher-order structure without changing amino-acid order.

Step 2: X permits folding before the covalent cross-links are re-established. Y locks links into an unfolded context, so removal of urea alone need not remove those constraints.

Step 3: Y's rescue by reduction and refolding shows that its low activity is not an irreversible sequence deletion. It supports a reversible structural trap in the supplied model.

Why: Function depends on higher-order organisation and suitable cross-links, not only the presence of the correct sequence.

Why not the others: (D) contradicts the sequencing and rescue. (B) confuses side-chain disulfides with backbone peptide bonds. (C) invents gene loss in a purified-protein manipulation.

Key point: Distinguish peptide sequence, noncovalent folding and disulfide constraints.

Q15 / B Mixed-substrate respiratory quotient resolved by two independent oxidation balances

Answer: (B) The glucose-associated oxygen fraction is 6/29.

Given: Total O2=29 mmol and CO2=22 mmol; the two substrates alone supply carbon.

Formula/Principle: Balance complete oxidation separately before combining respiratory quotients.

Step 1:

$$C_6H_{12}O_6 + 6O_2 \longrightarrow 6CO_2 + 6H_2O, \qquad C_{16}H_{32}O_2 + 23O_2 \longrightarrow 16CO_2 + 16H_2O.$$

Step 2: If the substrate amounts are g and f mmol,

$$6g + 23f = 29, \qquad 6g + 16f = 22.$$

Subtraction gives f=1, then g=1.

Step 3: Glucose accounts for 6 mmol of the consumed oxygen. Its fraction is

$$6/29.$$

Why: The overall respiratory quotient is oxygen-weighted; equal substrate moles do not imply equal oxygen contributions.

Why not the others: (A) is the overall CO2/O2 ratio, not the requested contribution. (C) treats equal substrate amounts as equal oxygen use. (D) uses fatty-acid CO2 production as if it were glucose oxygen consumption.

Key point: Resolve mixed respiration with both carbon and oxygen balances.

Chemistry

Q16 / C Geometrical, dilution and donor-masking controls distinguish internal hydrogen bonding from a liquid network

Answer: (C) X matches the ortho geometry P and Y the para geometry Q; the controls distinguish internal from intermolecular hydrogen bonding.

Step 1: Adjacent OH and nitro oxygen in P permit an internal donor–acceptor contact. Opposite positions in Q do not provide the corresponding short internal contact.

Step 2: Dilution separates molecules and disrupts intermolecular association without necessarily breaking a within-molecule interaction. The masking control removes the OH donor, supporting the specific role of that group rather than a general mass effect.

Step 3: Reduced intermolecular networking helps explain X being more volatile; equal monomer masses in vapour do not make the liquid interactions equal.

Why: Geometry, concentration dependence and removal of a donor are independent constraints. No dimer fraction or covalent polymerisation follows from these qualitative controls.

Why not the others: (A) reverses the positional assignment and the concentration argument. (B) confuses equal molecular mass with equal liquid association. (D) mistakes reversible noncovalent interaction for a changed molecular formula.

Key point: Test an interaction through geometry and controlled perturbation, not molar mass alone.

Q17 / A Synergic carbonyl donation strengthens one bond while weakening another

Answer: (A) Stronger back-donation reinforces metal–carbon bonding and weakens the internal CO bond.

Why: The two interactions are complementary: sigma donation from CO helps establish the metal–ligand connection, and metal-to-ligand pi back-donation adds another bonding contribution between those partners. The receiving CO orbital is antibonding with respect to its own C–O bond, so increasing occupancy there lowers the internal bond order. Metal–ligand strength and intraligand strength can therefore change in opposite directions.

Why not the others: (B) ignores the bonding role of the metal–ligand overlap. (C) reverses the consequence of occupying an antibonding orbital. (D) denies the donor interaction specified in the question and present in the carbonyl model.

Key point: Identify which bond an orbital is bonding or antibonding with respect to.

Q18 / D Fusion threshold followed by shared water-and-vessel sensible heating

Answer: (D) All ice melts and the final temperature is 660/92≈7.17 degrees C.

Given: The ice and vessel start at the melting temperature.

Formula/Principle: Supply latent heat before sensible heating, including the vessel only when temperature rises.

Step 1: Complete melting requires

$$Q_{fus} = 10.0(334) = 3340\ \text{J}.$$

The 4000 J input exceeds this.

Step 2: The remaining 660 J warms water and vessel together, whose combined heat capacity is

$$C = 10.0(4.20) + 50 = 92\ \text{J K}^{-1}.$$

Step 3:

$$T_f = 0 + 660/92 \approx 7.17\ ^\circ\text{C}.$$

Why: The energy budget crosses a phase boundary; neither one heat-capacity division nor latent heat alone describes the complete process.

Why not the others: (A) ignores that input exceeds complete fusion heat. (B) omits latent heat. (C) omits vessel heating after melting.

Key point: Account for every phase and every body whose temperature changes.

Q19 / B Reduction conditions must remove a ketone oxygen while preserving an acid-sensitive acetal

Answer: (B) Wolff–Kishner conditions reduce the ketone while preserving the base-stable acetal.

Why: The ketone first condenses with hydrazine to form a hydrazone. Strong base and heat then convert that group into a methylene group with loss of nitrogen. The separate acetal is stable to these basic conditions; it is normally hydrolysed back to a carbonyl compound in aqueous acid. Thus the choice must satisfy both the desired reduction and the survival of the protecting group. Clemmensen reduction can remove a ketone oxygen, but its strongly acidic aqueous medium threatens the acetal, so choosing a reduction name alone is insufficient.

Why not the others: (A) reverses acetal stability: Zn(Hg)/HCl is acidic and can deprotect it. (C) NaBH4 gives an alcohol rather than a methylene group, and the later acid also threatens the acetal. (D) Acid hydrolyses an acetal but supplies no ketone-to-methylene reduction.

Key point: Select reaction conditions using both the target transformation and the stability of every retained functional group.

Q20 / C Scratches separate sacrificial-zinc protection from a noble-metal galvanic coating

Answer: (C) Zinc can protect sacrificially, whereas copper can promote anodic iron dissolution at a scratch.

Step 1: Zinc's more negative reduction potential makes its oxidation favourable relative to iron in this comparison. Connected zinc can supply electrons while iron acts as the protected cathodic partner.

Step 2: Copper is more noble than iron. Exposed iron coupled to copper can instead become the anode and dissolve.

Why: An intact barrier and a galvanically connected scratched coating are different protection situations. The potential order determines which metal sacrifices itself; the mere presence of a metallic layer does not fix that order.

Why not the others: (A) ignores the exposed galvanic junction. (B) reverses the supplied potential order. (D) infers electrode role from geometrical position instead of electrochemical tendency.

Key point: Reassess coating protection after the substrate becomes exposed.

Q21 / B Two amide-to-amine routes retain different labelled carbons before a common deamination

Answer: (B) P gives unlabelled methanol and Q gives ethanol with 13C at the OH-bearing carbon.

Step 1: Hofmann bromamide degradation removes the amide carbonyl carbon from the organic amine. P therefore gives CH3NH2, with the label excluded from that one-carbon product.

Step 2: Ordinary aliphatic primary-amine deamination replaces NH2 by OH with nitrogen evolution, giving unlabelled CH3OH.

Step 3: Hydride reduction retains the amide carbonyl carbon as the amine-bearing methylene, giving CH3–13CH2–NH2 in Q. Deamination then gives CH3–13CH2–OH.

Why: The two amide-to-amine methods have different carbon-retention rules; the common later step preserves that difference. Tracking the labelled atom through each reaction is necessary, not just counting the original total carbons.

Why not the others: (A) exchanges the two preparation methods. (C) retains a carbon lost in P and moves the Q label to the wrong carbon. (D) incorrectly applies Hofmann carbon loss to hydride reduction.

Key point: Track the amide carbonyl carbon before applying a later common reaction.

Q22 / B Mixed-valence iron accounting precedes selective permanganate demand

Answer: (B) The volume is 100 mL and final Fe3+ amount is 0.0300 mol.

Given: Fe3O4 contains one Fe(II) and two Fe(III) centres per formula unit.

Formula/Principle: Only the Fe(II) fraction supplies electrons; one acidified permanganate ion accepts five electrons.

Step 1:

$$n_{oxide} = 2.32/232 = 0.0100 \text{ mol}, \quad n_{Fe^{2+}} = 0.0100 \text{ mol}.$$

Step 2:

$$MnO_4^- + 5Fe^{2+} + 8H^+ \longrightarrow Mn^{2+} + 5Fe^{3+} + 4H_2O.$$

Thus titrant moles are 0.0100/5=0.00200.

Step 3:

$$V = 0.00200/0.0200 = 0.100 \text{ L}; \qquad n_{Fe^{3+},final} = 0.0200 + 0.0100 = 0.0300 \text{ mol}.$$

Why: The average iron oxidation number is not a population of identical iron ions for the selective redox assay.

Why not the others: (A) treats all iron as Fe(II). (C) omits initially present Fe(III). (D) divides the volume by five a second time.

Key point: Resolve mixed valence before counting titratable electrons.

Q23 / D Species disappearance rate differs from the normalised reaction rate and product concentration rise

Answer: (D) Average reaction rate is 0.00100 M s^{-1} and B rises by 0.0900 M.

Given: A decreases by 0.0600 M in 30.0 s and the reaction coefficients are 2 and 3.

Formula/Principle: $r_{avg} = -\frac{1}{2}\Delta[A]/\Delta t = \frac{1}{3}\Delta[B]/\Delta t$.

Step 1: A disappearance rate is 0.0600/30.0=0.00200 M s^{-1}, so

$$r_{avg} = 0.00200/2 = 0.00100 \text{ M s}^{-1}.$$

Step 2: The stoichiometric extent in concentration units is 0.0600/2=0.0300 M, giving

$$\Delta[B] = 3(0.0300) = 0.0900 \text{ M}.$$

Why: Different species have different appearance or disappearance rates, while division by their stoichiometric coefficients defines one common reaction rate. One interval does not determine reaction order.

Why not the others: (A) reports A disappearance instead of the normalised rate. (B) assumes a 1:1 A-to-B conversion. (C) makes both errors.

Key point: Normalise the rate and conserve stoichiometric amounts separately.

Q24 / A Distance-dependent induction reverses the corresponding carboxylate basicity order

Answer: (A) The nearer electron-withdrawing chlorine enhances acidity more, reversing the conjugate-base basicity order.

Why: Chlorine withdraws electron density through sigma bonds. Its stabilising influence on a nearby carboxylate is strongest when closest to the acid group and decreases with distance along the chain. More stable carboxylate corresponds to easier acid dissociation, so the C2-substituted acid is strongest in this comparison. The conjugate base of a stronger acid is correspondingly weaker under the same conditions. Positional isomers need not have identical acidity even though their elemental formulas agree.

Why not the others: (B) reverses the distance dependence for acidity. (C) treats composition as a complete description of electronic effects. (D) makes conjugate-base basicity increase with its parent acid's acidity.

Key point: Compare both substituent distance and the direction of the conjugate-acid/base relation.

Q25 / C Functional-group priority and double-bond configuration combined in one unsaturated hydroxy acid

Answer: (C) (E)-4-Hydroxybut-2-enoic acid.

Step 1: Carboxylic acid is the principal functional group, so its carbon is C1 and the four-carbon parent is butenoic acid. The double bond lies between C2 and C3; the OH-bearing terminal carbon is C4.

Step 2: On each alkene carbon the carbon-containing group has priority over hydrogen. The given opposite placement is E.

Why: Functional-group priority determines the parent and numbering before the stereodescriptor is assigned. Merely numbering from the alcohol end would give a plausible-looking but inconsistent name.

Why not the others: (A) assigns the opposite alkene configuration. (B) puts the hydroxyl on the acid carbon rather than on the terminal CH2 group. (D) makes the alcohol the principal suffix and changes the parent description.

Key point: Determine principal group, locants and configuration in that order.

Q26 / A Volatility invalidates a simplified total-vapour inference but not a sealed retained-solute osmotic measurement

Answer: (A) The osmotic result is usable under the stated membrane condition; total vapour pressure requires both components.

Given: The vessel is sealed and the membrane excludes the solute, preventing loss or leakage from invalidating the stated concentration.

Formula/Principle: $\pi = cRT$ for the dilute molecular solution, while volatile-component total pressure is the sum of both partial pressures.

Step 1:

$$c = \pi/(RT) = 0.100 \text{ mol L}^{-1}.$$

Step 2:

$$M = \frac{10.0 \text{ g L}^{-1}}{0.100 \text{ mol L}^{-1}} = 100 \text{ g mol}^{-1}.$$

Why: The nonvolatile-solute lowering formula assumes the solute does not itself supply vapour pressure. That assumption is distinct from solute exclusion by an osmotic membrane.

Why not the others: (B) converts failure of an inappropriate vapour formula into an impossible mass. (C) ignores the explicit retention and sealed-system assumptions. (D) neither follows the osmotic division nor includes solute vapour.

Key point: Check the separate physical assumptions of two colligative measurements.

Q27 / C Impure oxidant and combustion products jointly fix empirical ratio and excess oxygen

Answer: (C) CH4 and 0.0200 mol oxygen remaining.

Given: The oxidant's stated total includes inert gas.

Formula/Principle: Product atom balances determine empirical composition and actual oxygen consumption.

Step 1: Carbon is 0.0500 mol and hydrogen atoms are 2(0.100)=0.200 mol, giving C:H=1:4. Their masses sum to $0.0500(12) + 0.200(1) = 0.800$ g, checking the sample.

Step 2: Product oxygen demand is

$$n_{O_2} = n_{CO_2} + \tfrac{1}{2} n_{H_2O} = 0.0500 + 0.0500 = 0.100 \text{ mol}.$$

Step 3: Available oxygen is $0.150(0.800) = 0.120$ mol, leaving

$$0.120 - 0.100 = 0.0200 \text{ mol}.$$

Why: Oxygen is in excess and the organic sample is completely consumed. Counting inert feed as oxygen would overstate the excess.

Why not the others: (A) forgets the two hydrogen atoms per water molecule. (B) treats the entire feed as oxygen. (D) does not reduce the empirical ratio and ignores oxygen consumption.

Key point: Separate feed composition from reaction stoichiometry.

Q28 / B The argon–potassium mass inversion tests the actual variable in the modern periodic law

Answer: (B) Atomic numbers 18 then 19 fix argon before potassium.

Why: The modern periodic law relates periodic properties to atomic number. Relative atomic mass is an abundance-weighted isotopic property and need not increase in the same order for every adjacent pair. Argon–potassium illustrates why a mass-only ordering cannot replace the nuclear-charge ordering. Isotopes of the same element retain its atomic number and therefore do not receive separate periodic-table positions.

Why not the others: (A) reapplies an atomic-mass criterion instead of the modern law. (C) treats approximate mass equality as equality of element identity. (D) uses neutron count where nuclear charge is required.

Key point: Keep elemental ordering distinct from isotopic mass.

Q29 / D Vitamin solubility predicts absorption vulnerability without an absolute rule about all water-soluble stores

Answer: (D) Solubility influences absorption and storage, but does not impose one universal deficiency timetable.

Why: Vitamins A,D,E and K are fat soluble and can be affected when fat absorption is impaired. Vitamin C is water soluble and is not stored in the same way as typical fat-soluble reserves, so inadequate intake may produce a different time course. Water solubility must not be turned into an exceptionless statement that every B vitamin has no significant stores; vitamin B12 is an important storage exception. The question concerns the supported class distinction rather than diagnosing a specific patient from duration alone.

Why not the others: (A) removes meaningful absorption and storage differences. (B) classifies chemistry from a name. (C) reverses the solubility-related absorption vulnerability.

Key point: Use solubility as a biological constraint while retaining known exceptions.

Q30 / A Both conjugate-ion hydrolyses determine the pH of a weak-acid weak-base salt

Answer: (A) The salt is acidic, with approximate pH 6.

Given: Parent-base Kb=10^{-6} and parent-acid Ka=10^{-4}.

Formula/Principle: Both ions hydrolyse; $pH = 7 + \frac{1}{2}\log(K_b/K_a)$ under the specified approximation.

Step 1: Their separate constants are

$$K_a(BH^+) = K_w/K_b = 10^{-8}, \quad K_b(A^-) = K_w/K_a = 10^{-10}.$$

The cation's acidic action is stronger.

Step 2:

$$pH = 7 + \tfrac{1}{2}\log(10^{-6}/10^{-4}) = 7 - 1 = 6.$$

Why: Equal stoichiometric charges do not mean equal acid and base strengths. Both parent equilibria, not only the 1:1 salt formula, determine hydrolysis.

Why not the others: (B) confuses electrical neutrality with neutral pH. (C) reverses the conjugate-strength relation and the logarithmic ratio. (D) treats weak hydrolysis as complete strong-acid dissociation.

Key point: Recover both conjugate-ion constants before interpreting a weak/weak salt.

Mathematics

Q31 / A Parallel tangency and a horizontal-width parabolic segment integral

Answer: (A) The contact point is (4,4), and the segment area is 8/3.

Given: Both endpoints lie on the parabola's upper branch.

Formula/Principle: Match the derivative to the chord slope, then integrate horizontal widths over the chord's y range.

Step 1: Chord slope is (6-2)/(9-1)=1/2. From 2y y'=4, y'=2/y, so parallel tangency has y=4 and x=4.

Step 2: The chord is x=2y-3. Between y=2 and 6 it lies to the right of $x=y^2/4$.

Worked derivation:

$$A = \int_2^6 (2y - 3 - y^2/4)\, dy = \left[y^2 - 3y - y^3/12\right]_2^6 = 8/3.$$

Why: The tangent condition and the area use the same chord but different geometric relations. Horizontal widths avoid incorrectly expressing both branches as one y(x) boundary.

Why not the others: (B) halves the segment area. (C) gives a point not on $y^2=4x$. (D) doubles the area.

Key point: Choose the integration direction that gives one interval and one pair of boundaries.

Q32 / D Outer stationary preimages and an inner derivative zero contribute different extrema

Answer: (D) Zero is a local maximum, and both -1 and 1 are local minima.

Given: The outer derivative is supplied; an explicit formula for f is unnecessary.

Formula/Principle: Apply the chain rule and test signs on the real intervals separated by stationary points.

Step 1:

$$g'(x) = 2x\frac{x^2 - 1}{x^2 + 1}.$$

Its denominator is positive and its zeros are -1,0,1.

Step 2: The derivative signs are negative, positive, negative, positive on the four successive intervals.

Worked derivation: The - to + changes at ±1 give minima; the + to - change at zero gives a maximum.

Why: A stationary composite can arise from an outer derivative zero or from the inner derivative zero. Ignoring the latter loses x=0.

Why not the others: (A) misses two outer-derivative preimages and reverses zero's type. (B) misses zero and reverses the other types. (C) reverses the number of maxima and minima.

Key point: Include every factor of the composite derivative before classifying sign changes.

Q33 / B Two integration-by-parts terms and endpoint accounting after translation

Answer: (B) (5-6 ln 2)/16.

Given: The denominator is positive throughout the integration interval.

Formula/Principle: Translate the variable and split the rational factor before using integration by parts.

Step 1: Set u=1+x. The integral becomes

$$I = \int_1^2 \frac{\ln u}{u^2}\, du - \int_1^2 \frac{\ln u}{u^3}\, du.$$

Step 2: Parts gives

$$I_1 = \left[-\frac{\ln u + 1}{u}\right]_1^2 = (1 - \ln 2)/2, \quad I_2 = \left[-\frac{\ln u}{2u^2} - \frac{1}{4u^2}\right]_1^2 = 3/16 - \ln 2/8.$$

Worked derivation:

$$I = I_1 - I_2 = 5/16 - 3\ln 2/8 = (5 - 6\ln 2)/16.$$

Why: Translating changes the numerator x into u-1; retaining both terms is essential to the endpoint cancellation.

Why not the others: (A) keeps only I1. (C) undercounts the logarithmic term in the subtraction. (D) reports I2 alone.

Key point: Split the translated numerator before integrating the logarithmic factors.

Q34 / C Counting symmetric transitive relations uses inactive elements and vacuous cases rather than adding closure pairs

Answer: (C) 10.

Given: An empty relation is permitted; reflexivity must hold on every element of the stated set to count as reflexive.

Formula/Principle: For a symmetric transitive relation, every element appearing in a pair must relate to itself. The active elements form equivalence classes; the remaining elements appear in no pairs.

Step 1: If (a,b) occurs, symmetry gives (b,a), then transitivity gives (a,a) and (b,b). Thus nonreflexivity means at least one inactive element.

Step 2: With zero active elements there is one empty relation. With one active element there are three choices. With two active elements there are three subsets, each allowing either two singleton classes or one combined class.

Worked derivation:

$$N = 1 + 3 + \binom{3}{2}(2) = 10.$$

Why: All-active equivalence relations are excluded, but the empty relation is symmetric and transitive vacuously and must be retained.

Why not the others: (A) counts only one-active-element cases. (B) counts only two-active-element cases. (D) includes the five reflexive equivalence relations on all three elements.

Key point: Use the set of active elements when a relation is not required to be reflexive.

Q35 / A A dependent row constrains consistency before the solution family is counted

Answer: (A) Consistency requires a=6, and a free parameter remains.

Given: The second coefficient row is twice the first.

Formula/Principle: Row dependence must be matched by the same dependence in the right-hand side.

Step 1: Doubling the first equation makes its right side 6, so consistency requires a=6.

Step 2: With z=1, the remaining independent equation is x+2y=2.

Worked derivation:

$$(x, y, z) = (2 - 2t, t, 1), \qquad t \in \mathbb{R}.$$

Why: A singular matrix prevents a unique solution but allows either no solutions or infinitely many, depending on consistency of the right-hand side.

Why not the others: (B) fails the row-dependence condition. (C) discards the free y parameter. (D) confuses noninvertibility with unavoidable inconsistency.

Key point: Compare dependent rows and their right-hand sides together.

Q36 / B A second-failure stopping condition reweights unequal-probability prefixes

Answer: (B) 5/11.

Given: Stopping at trial 4 means exactly one failure among the first three and failure on trial 4.

Formula/Principle: Enumerate the allowed prefixes with their unequal independent probabilities; the fourth-trial failure factor cancels conditionally.

Step 1: Prefix probabilities are

$$P(FSS) = 1/4, \quad P(SFS) = 1/8, \quad P(SSF) = 1/12.$$

Step 2: Only the last two have a first-trial success.

Worked derivation:

$$P(S_1 \mid T = 4) = \frac{1/8 + 1/12}{1/4 + 1/8 + 1/12} = \frac{5/24}{11/24} = 5/11.$$

Why: The stopping condition constrains the prefix but does not make its three allowed arrangements equally probable.

Why not the others: (A) counts two of three patterns without weighting them. (C) ignores the information in the stopping event. (D) uses an incorrect numerator or excludes an allowed success prefix.

Key point: A stopping event selects weighted histories rather than uniformly selected patterns.

Q37 / D Alternating digit sums and leading-zero exclusion constrain a full digit permutation

Answer: (D) 16.

Given: Zero cannot occupy the leading position.

Formula/Principle: For divisibility by 11, the alternating digit sums differ by a multiple of 11.

Step 1: Total digit sum is 10. If the three odd-position digits sum to S, the difference is 2S-10. Its possible range excludes nonzero multiples of 11, so S=5.

Step 2: The odd-position triples are {0,1,4} and {0,2,3}. Each admits 3!2! arrangements before the leading restriction.

Step 3: In either triple, placing zero first leaves 2!2! invalid arrangements.

Worked derivation:

$$N = 2(3!2! - 2!2!) = 2(12 - 4) = 16.$$

Why: Selecting an alternating-position subset and preventing leading zero are separate constraints.

Why not the others: (A) retains leading-zero strings. (B) counts only one uncorrected triple. (C) removes too few leading-zero arrangements.

Key point: Apply divisibility to position sets before arranging their digits.

Q38 / A One shared binomial coefficient fixes an integer exponent but not a later coefficient

Answer: (A) 20-5√5.

Given: n is an integer at least four, so the coefficient formulas and exponent restrictions apply.

Formula/Principle: Compare the same power using the binomial coefficient and its base factor before solving for n.

Step 1:

$$\binom{n}{2} = 5\binom{n-3}{2} \;\Rightarrow\; n(n-1) = 5(n-3)(n-4).$$

Step 2: This gives $2n^2 - 17n + 30 = 0$, with roots 6 and 5/2. Only n=6 is admissible.

Step 3: The desired coefficients are $\binom{6}{3} = 20$ and $\binom{3}{3}(\sqrt{5})^3 = 5\sqrt{5}$.

Worked derivation:

$$c_{3,first} - c_{3,second} = 20 - 5\sqrt{5}.$$

Why: Agreement of one coefficient does not make two binomial expansions identical; the radical factor is raised to the power of the term being compared.

Why not the others: (B) adds instead of subtracting. (C) forgets to cube √5. (D) reverses the subtraction order.

Key point: Enforce the integer exponent condition before comparing the next coefficients.

Q39 / C Odd-integer arithmetic and square sums determine an integer term count

Answer: (C) n=6.

Given: The terms are 1,3,...,2n-1.

Formula/Principle: Sum the arithmetic progression and expand the square using standard sums of integers and squares.

Step 1:

$$S_1 = n^2, \qquad S_2 = \sum_{k=1}^{n} (2k-1)^2 = \frac{n(4n^2-1)}{3}.$$

Step 2: The condition is S2=8S1-2, giving

$$4n^3 - 24n^2 - n + 6 = 0.$$

Worked derivation:

$$(n-6)(4n^2-1) = 0.$$

The other roots are ±1/2, leaving n=6 as the only positive integer.

Why: n counts terms and cannot be accepted merely as a positive real root. At six terms the sums are 36 and 286, satisfying 286=8(36)-2.

Why not the others: (A) fails the stated square-sum relation. (B) also fails it. (D) solves neither factor of the derived equation.

Key point: Preserve the term-count domain when solving an equation from special sums.

Q40 / B A unique removable extension is differentiable but cannot repair an oscillating neighbouring derivative

Answer: (B) The unique continuous extension is differentiable but its derivative is discontinuous.

Given: Oscillation is multiplied by x².

Formula/Principle: First test continuity, then the derivative definition, then the nearby derivative values.

Step 1: $|x^2 \sin(1/x)| \le x^2$ forces the limit zero, so continuity requires a=0.

Step 2: With that value,

$$f'(0) = \lim_{h \to 0} h \sin(1/h) = 0.$$

Step 3: For nonzero x,

$$f'(x) = 2x \sin(1/x) - \cos(1/x).$$

The first term tends to zero but the second has different limits along $x = 1/(2\pi n)$ and $x = 1/((2n+1)\pi)$.

Worked derivation: These derivative subsequences tend to -1 and +1, so f' has no limit at zero.

Why: Changing a point value can repair continuity of f but cannot alter derivative values at neighbouring nonzero points.

Why not the others: (A) stops after computing the derivative at the point. (C) ignores the bounded derivative quotient. (D) ignores the necessary value for continuity.

Key point: Differentiability at a point does not imply continuity of the derivative.

Q41 / D Separation loses a family of trigonometric equilibria unless checked first

Answer: (D) Constants are all integer multiples of π; the specified solution takes value 2π/3.

Given: The nonconstant branch remains between zero and π.

Formula/Principle: Check zero factors before dividing in separation; $\int \csc y\,dy = \ln|\tan(y/2)| + C$.

Step 1: Constant y requires sin y=0, giving y=nπ. These would be lost if one divided by sin y without checking.

Step 2: On the given branch separation and the initial condition give

$$\ln\tan(y/2) = x, \quad \tan(y/2) = e^x.$$

Worked derivation: At x=ln√3,

$$\tan(y/2) = \sqrt{3}, \quad y/2 = \pi/3, \quad y = 2\pi/3.$$

Why: The branch constraint selects the correct inverse tangent, while the equilibrium solutions must be listed separately from the divided equation.

Why not the others: (A) misses other equilibria and the branch value. (B) has the equilibria but the reciprocal angle. (C) misses all other constant multiples.

Key point: Recover constant solutions excluded by separation.

Q42 / A An oblique prescribed direction changes the point-plane connecting distance

Answer: (A) Q=(-2,2,0), and the directed-line distance is 3√2.

Given: The connecting direction is prescribed and is not the plane normal.

Formula/Principle: Parametrise the line, then impose the plane equation and measure its segment.

Step 1:

$$Q = (1+t, 2, 3+t), \quad (1+t) + 2 + (3+t) = 0 \;\Rightarrow\; t = -3.$$

Step 2: Q=(-2,2,0).

Worked derivation:

$$PQ = |-3|\sqrt{1^2+0^2+1^2} = 3\sqrt{2}.$$

Why: The distance along a specified oblique line is not generally the shortest perpendicular distance to the plane, which here is 2√3.

Why not the others: (B) uses the perpendicular-foot construction instead of the prescribed direction. (C) does not reach the plane. (D) combines the correct point with the wrong distance.

Key point: Distinguish a constrained segment from the perpendicular minimum.

Q43 / C A free parallel component in a cross-product equation is fixed by minimum norm

Answer: (C) The minimum squared magnitude is 2.

Given: A cross product does not determine the component of v parallel to a.

Formula/Principle: Solve the component equations and minimise the remaining quadratic, or choose the solution perpendicular to a.

Step 1: If v=(x,y,z),

$$a \times v = (z, -z, y-x) = (0, 0, 2).$$

Hence z=0,y=x+2, so v=(t,t+2,0).

Step 2:

$$|v|^2 = t^2 + (t+2)^2 = 2(t+1)^2 + 2.$$

Worked derivation: Minimum is 2 at t=-1, giving v=(-1,1,0), perpendicular to a.

Why: Adding an arbitrary parallel component leaves the cross product unchanged but increases the norm away from the perpendicular solution.

Why not the others: (A) falls below the established quadratic lower bound. (B) is the norm squared of the nonminimal solution (0,2,0). (D) is impossible for a nonzero required cross product.

Key point: Resolve the cross-product kernel before minimising length.

Q44 / B A grouped population variance gives two real roots but only one valid integer frequency

Answer: (B) a=2 and the midpoint-based mean is 13/5.

Given: Class representatives are 1,3,5 and total frequency is a+3.

Formula/Principle: Population variance is weighted $E[X^2]$-$E[X]^2$.

Step 1:

$$\bar{x} = (a+11)/(a+3), \quad E[X^2] = (a+43)/(a+3).$$

Step 2:

$$\sigma^2 = \frac{24a+8}{(a+3)^2} = 56/25 \;\Rightarrow\; 7a^2 - 33a + 38 = 0.$$

Step 3: Roots are 2 and 19/7. Only 2 is an integer frequency.

Worked derivation:

$$\bar{x} = (2+11)/(2+3) = 13/5.$$

Why: A real solution to the variance equation is not automatically an admissible count; the population divisor also changes with the unknown frequency.

Why not the others: (A) retains a noninteger frequency. (C) computes the unweighted midpoint mean. (D) gives variance 2 rather than 56/25.

Key point: Apply count integrality after solving the weighted variance equation.

Q45 / D A principal cosine inverse produces two distinct signed branches before inverse-angle values are summed

Answer: (D) 2π/9.

Given: The inverse functions require -1≤x≤1. Put θ=arcsin x, so -π/2≤θ≤π/2.

Formula/Principle: Since $2x^2 - 1 = -\cos 2\theta$, the principal cosine inverse is $\pi - 2|\theta|$.

Step 1: The equation becomes

$$\theta + \pi - 2|\theta| = 2\pi/3.$$

Step 2: For θ≥0 it gives π-θ=2π/3, hence θ=π/3. For θ<0 it gives π+3θ=2π/3, hence θ=-π/9. Both values belong to their assumed branches.

Step 3: Arcsin is one-to-one on its principal range, so the two θ values correspond to two distinct x solutions.

Worked derivation:

$$\sum \arcsin x = \pi/3 - \pi/9 = 2\pi/9.$$

Why: The absolute-value fold produces two different linear equations. Solving only the nonnegative-angle branch loses a valid negative solution.

Why not the others: (A) keeps only the positive branch. (B) keeps only the negative branch. (C) adds the magnitude of the negative angle rather than its signed value.

Key point: Resolve both principal-value branches before summing over solutions.

Physics

Q46 / C Subtractive cancellation controls precision in a later density quotient

Answer: (C) 0.64 g cm⁻³ with two significant figures.

Given: Both masses are measured to 0.001 g and volume has three significant figures.

Formula/Principle: Preserve the allowed decimal place through subtraction before deciding the quotient's significant figures.

Step 1:

$$\Delta m = 20.600 - 20.584 = 0.016 \text{ g}.$$

This difference has two significant figures.

Step 2:

$$\rho = 0.016/0.0250 = 0.640 \text{ g cm}^{-3}.$$

Worked derivation: The two-significant-figure numerator limits the reported result to 0.64 g cm^{-3}.

Why: Cancellation changes the number of meaningful significant digits; rounding the two large masses first can erase the measured difference.

Why not the others: (A) reports unsupported precision. (B) discards an allowed digit. (D) substitutes prematurely rounded inputs for the original measurements.

Key point: Follow operation-specific precision rules in their actual order.

Q47 / B Opposite static-friction directions produce lower and upper banked-road speed limits

Answer: (B) The permitted interval is 10 to 20 m s⁻¹.

Given: Radius r=20 m, bank angle θ=45 degrees and μ=1/3.

Formula/Principle: At each limiting speed static friction is μN, with opposite directions at the two limits.

Step 1: At maximum speed friction acts down the bank:

$$N\sin\theta + \mu N\cos\theta = mv_{max}^2/r, \quad N\cos\theta - \mu N\sin\theta = mg.$$

Step 2: Their ratio gives

$$v_{max}^2 = rg\frac{\tan\theta + \mu}{1 - \mu\tan\theta} = 200(2) = 400.$$

Step 3: At minimum speed friction acts up the bank, giving

$$v_{min}^2 = rg\frac{\tan\theta - \mu}{1 + \mu\tan\theta} = 200(1/2) = 100.$$

Worked derivation:

$$v_{min} = 10, \quad v_{max} = 20 \text{ m s}^{-1}.$$

Why: At low speed friction prevents inward/down-bank slip; at high speed it helps provide extra inward force. One friction direction cannot describe both limits.

Why not the others: (A) omits the low-speed slip limit. (C) incorrectly treats the frictionless design speed as the upper limit. (D) excludes the range made possible by static friction.

Key point: Derive both limiting force balances before reporting a banked-road range.

Q48 / A Longitudinal extension and transverse contraction jointly change an elastic wire's resistance

Answer: (A) Force 2000 N and first-order resistance 10.2 ohm.

Given: Engineering stress uses the original area in this linear small-strain model.

Formula/Principle: F=YAε and R=ρL/A; constant volume makes fractional area change approximately -ε.

Step 1:

$$F = (2.0 \times 10^{11})(1.00 \times 10^{-6})(0.010) = 2000 \text{ N}.$$

Step 2: Length grows by ε while area decreases by ε to first order, so

$$R'/R \simeq 1 + 2\varepsilon = 1.020.$$

Worked derivation:

$$R' \simeq 10.0(1.020) = 10.2\ \Omega.$$

Why: Both geometrical changes increase resistance. The model consistently neglects second-order strain terms rather than mixing an exact volume correction with finite-strain elasticity.

Why not the others: (B) ignores transverse contraction. (C) converts area or strain with a factor-ten error. (D) reverses the net resistance change.

Key point: Specify the approximation order when coupling elastic and electrical changes.

Q49 / D Removal of one parallel heat-loss channel shifts the steady radiative temperature

Answer: (D) Initial radiation is 4 W, and removal of conduction requires a higher steady temperature.

Given: Input power is constant and only the specified heat-loss channels operate.

Formula/Principle: At steady state total outward power equals inward power.

Step 1:

$$P_{rad}(T_0) = 10 - 6 = 4 \text{ W}.$$

Step 2: Without conduction, at T0 only 4 W escapes while 10 W enters, so internal energy initially rises. New equilibrium requires radiation alone to reach 10 W.

Worked derivation: Since $P_{rad}(T)$ is increasing, $P_{rad}(T_{new}) = 10 > 4 = P_{rad}(T_0)$ implies $T_{new} > T_0$.

Why: Removing a heat-loss route does not remove the energy input; another route must carry more power at the new equilibrium.

Why not the others: (A) confuses the two original losses. (B) violates energy balance at the retained temperature. (C) ignores the original conductive power.

Key point: Evaluate the initial power imbalance before predicting the new thermal state.

Q50 / B Equal thermodynamic states need not have equal laboratory molecular energy when one gas drifts

Answer: (B) The laboratory mean kinetic-energy ratio is 2.

Given: Pressure and number density refer to random thermal motion in each packet's own rest frame.

Formula/Principle: p=nkT and the mean square of bulk plus zero-mean random velocity is U²+w².

Step 1: Equal p/n gives equal T and therefore equal thermal w² for the same gas. A has $\langle v_A^2 \rangle = w^2$.

Step 2: For B, the cross term vanishes because the random velocity has zero mean:

$$\langle v_B^2 \rangle = U^2 + w^2 = 2w^2.$$

Worked derivation:

$$\frac{\langle K_B \rangle}{\langle K_A \rangle} = \frac{m(2w^2)/2}{mw^2/2} = 2.$$

Why: Thermal pressure fixes random-motion energy, not the additional energy of uniform bulk translation in a chosen laboratory frame.

Why not the others: (A) omits bulk kinetic energy. (C) reports a speed ratio instead of an energy ratio. (D) adds rms speeds as collinear deterministic velocities before squaring.

Key point: Separate thermal motion from the velocity of the gas as a whole.

Q51 / C A delayed same-frequency contribution changes the resultant SHM amplitude through phase

Answer: (C) The resultant amplitude is A.

Given: Equal frequencies and amplitudes, with a start-time phase difference ωτ=2π/3.

Formula/Principle: Add same-frequency displacements as phasors or use the cosine sum identity.

Step 1:

$$x_1 + x_2 = 2A\cos(\omega\tau/2)\cos(\omega t - \omega\tau/2).$$

Step 2: Since $\omega\tau/2 = \pi/3$, the prefactor is $2A(1/2) = A$.

Worked derivation:

$$A_{resultant} = |2A\cos(\pi/3)| = A.$$

Why: The delayed maximum of the second oscillator is not simultaneous with the first maximum. Amplitudes therefore cannot be added as unsigned scalars.

Why not the others: (A) assumes zero phase difference. (B) uses a phase difference π/3 instead of 2π/3. (D) assumes antiphase π.

Key point: Convert a delay into phase before adding oscillations.

Q52 / A Two different end reflections determine pulse sign and successive return times

Answer: (A) The free-end return is upward; the following fixed-end return is downward.

Given: The pulse starts halfway between two different boundary types.

Formula/Principle: Displacement reflection is noninverting at a free end and inverting at a fixed end; travel time is path length divided by speed.

Step 1: Travel midpoint→right end→midpoint has total length L. The free-end reflection retains the upward sign.

Step 2: The next leg midpoint→left end→midpoint adds another distance L. Reflection at the fixed end inverts the pulse.

Worked derivation:

$$t_1 = L/v, \quad t_2 = 2L/v, \quad (sign_1, sign_2) = (+, -).$$

Why: Boundary displacement constraints set phase independently of propagation direction. Reversing direction alone does not imply inversion.

Why not the others: (B) swaps free and fixed reflection rules. (C) uses one-way distances instead of complete return paths. (D) ignores the second fixed-end inversion.

Key point: Apply the correct boundary rule at every encounter.

Q53 / D A fixed Gaussian surface changes flux when charges exchange inside and outside positions

Answer: (D) The flux change is -3q/ε0.

Given: Charge positions relative to the surface change while its area does not.

Formula/Principle: Net outward electric flux is enclosed charge divided by ε0.

Step 1:

$$\Phi_X = q/\varepsilon_0, \qquad \Phi_Y = -2q/\varepsilon_0.$$

Step 2:

$$\Delta\Phi = \Phi_Y - \Phi_X = -3q/\varepsilon_0.$$

Worked derivation: The same signed result follows from +q leaving and -2q entering: $\Delta Q_{inside} = -q - 2q = -3q$.

Why: External charges can change the field on parts of a surface without adding net enclosed charge. Surface area alone does not fix the flux.

Why not the others: (A) reverses final minus initial. (B) combines the two charge values without tracking which side contains them. (C) assumes unchanged geometry implies unchanged flux.

Key point: Track signed charge crossing the boundary, not only surface area.

Q54 / B Potentiometer null removes tested-cell loading from an emf comparison

Answer: (B) E1/E2=2/3; the tested-cell current vanishes at each null.

Given: Potential gradient is fixed between the two comparisons.

Formula/Principle: At balance E=kℓ because no current is drawn from the test branch.

Step 1:

$$E_1/E_2 = (k \cdot 60)/(k \cdot 90) = 2/3.$$

Step 2: A test-cell internal drop is Ir; at a true null I=0 for that branch, so its internal resistance does not lower the balanced emf.

Worked derivation:

$$I_1 r_1 = I_2 r_2 = 0$$

without requiring r1=r2.

Why: A null comparison differs from a finite-resistance voltmeter that draws current and can measure a loaded terminal voltage.

Why not the others: (A) inverts the length ratio and overgeneralises resistance irrelevance. (C) ignores different balance lengths. (D) contradicts the null condition.

Key point: Identify the branch whose current is zero.

Q55 / C An inductor can return energy with unchanged current direction and reversed terminal voltage

Answer: (C) Current remains positive, voltage is negative and stored energy is falling.

Given: The terminal voltage uses the passive convention, with current entering A.

Formula/Principle: $V_A - V_B = L\,dI/dt$ and $dU/dt = LI\,dI/dt$.

Step 1: At ωt=3π/4,

$$I = I_0/\sqrt{2} > 0, \quad dI/dt = -\omega I_0/\sqrt{2} < 0.$$

Step 2:

$$V_A - V_B = -L\omega I_0/\sqrt{2} < 0.$$

Worked derivation:

$$dU/dt = -L\omega I_0^2/2 < 0.$$

Why: Voltage sign responds to the change of current, not to current sign alone. Here the inductor returns magnetic energy while current still flows in its original direction.

Why not the others: (A) assigns the increasing-current signs. (B) confuses negative slope with negative current. (D) uses the current maximum time instead of the stated time.

Key point: Distinguish current, current derivative and energy-flow signs.

Q56 / D Loaded transformer terminal voltage and efficiency include winding and core losses

Answer: (D) Load power is 64 W and efficiency 64 percent.

Given: Secondary induced emf is not identical to loaded terminal voltage because its winding resistance is retained.

Formula/Principle: Find secondary current through winding plus load, then include all stated losses in input power.

Step 1:

$$E_s = 100/5 = 20 \text{ V}, \quad I_s = 20/(1+4) = 4 \text{ A}.$$

Step 2:

$$P_{load} = I_s^2(4) = 64 \text{ W}, \quad P_{Cu} = I_s^2(1) = 16 \text{ W}.$$

Step 3: Input power supplies load plus copper and core losses.

Worked derivation:

$$\eta = \frac{64}{64+16+20} = 0.64.$$

Why: The turns relation applies to the induced emf in this model, while the load voltage is reduced by the secondary winding drop. Efficiency counts core loss in addition to copper loss.

Why not the others: (A) treats induced emf as load voltage. (B) omits the core loss from input. (C) calls the input power the delivered load power.

Key point: Separate induced emf, terminal voltage and the full power budget.

Q57 / A Eyepiece translation changes telescope image distance and angular magnification

Answer: (A) The final image distance is -20 cm and angular magnification magnitude is 20.

Given: The objective still forms its real image 80 cm behind it; the moved eyepiece is 84 cm behind it.

Formula/Principle: Apply the eyepiece lens formula and compare the output angle with the original distant-object angle.

Step 1: Eyepiece object distance is u=-4 cm.

$$1/5 = 1/v - 1/(-4) \;\Rightarrow\; 1/v = -1/20, \quad v = -20 \text{ cm}.$$

Step 2: If intermediate height is h, its virtual image height is $(v/u)h = 5h$. The output angle magnitude is 5h/20=h/4.

Worked derivation:

$$|M| = (h/4)/(h/80) = 20.$$

Why: Moving the eyepiece changes the finite-image setting but does not move the objective's intermediate image for the same distant object.

Why not the others: (B) gives a real instead of virtual final image. (C) retains normal-adjustment magnification. (D) reports the eyepiece object distance as its image distance.

Key point: Locate the unchanged intermediate image before applying the moved eyepiece.

Q58 / C Coincident diffraction zeros remove interference orders at the central-envelope boundary

Answer: (C) The third order is first missing and five interference maxima lie inside the central envelope.

Given: The diffraction envelope's first minima occur at |a sinθ|=λ.

Formula/Principle: A nominal interference maximum disappears when it coincides with a diffraction minimum.

Step 1: Equating the two angular conditions gives

$$m\lambda/d = n\lambda/a \;\Rightarrow\; m = 3n.$$

The first nonzero missing order is |m|=3.

Step 2: Strictly within the central envelope, $|m| < d/a = 3$.

Worked derivation: Allowed bright orders are

$$-2, -1, 0, 1, 2,$$

giving five.

Why: The envelope boundary orders ±3 cannot be counted as bright because the individual-slit amplitude vanishes there.

Why not the others: (A) confuses d/a with another order relation. (B) counts the extinguished boundary maxima. (D) skips the first coincident diffraction zero.

Key point: Apply the interference condition together with the finite-slit envelope.

Q59 / B A fixed total energy makes wavelength-product minimisation different from equal-momentum allocation

Answer: (B) Equal kinetic energies minimise the wavelength product, and λA/λB=2 there.

Given: Total kinetic energy K is fixed and both shares are positive.

Formula/Principle: $\lambda = h/\sqrt{2mK_i}$; minimising the product is equivalent to maximising KA KB.

Step 1:

$$\lambda_A\lambda_B = \frac{h^2}{4m\sqrt{K_AK_B}}.$$

Step 2: Since

$$K_AK_B = K_A(K - K_A) = K^2/4 - (K_A - K/2)^2,$$

it is largest at equal shares.

Step 3: Equal energies give momentum ratio pB/pA=√(4m/m)=2.

Worked derivation:

$$\lambda_A/\lambda_B = p_B/p_A = 2, \quad (\lambda_A\lambda_B)_{min} = h^2/(2mK).$$

Why: Equal momentum would produce equal wavelengths but would not maximise the kinetic-energy product. These are different constraints.

Why not the others: (A) gives equal momenta rather than the required minimum. (C) assumes equal energy means equal momentum for unequal masses. (D) does not maximise the energy product.

Key point: Optimise under the stated fixed sum before imposing a wavelength comparison.

Q60 / D Donor ionisation changes majority carriers while fixed-ion compensation preserves bulk neutrality

Answer: (D) Mobile majority electrons coexist with compensating positive ionised donors.

Given: The donor supplies an extra electron but the initially neutral material has not received net external charge.

Formula/Principle: Track the donor's charge after ionisation and distinguish mobile carriers from fixed lattice ions.

Step 1: A donated electron contributes negative mobile charge, while its donor becomes a positively charged fixed centre.

Step 2: Bulk neutrality requires

$$n \simeq N_D^+$$

when the small hole concentration is neglected; more exactly $n = p + N_D^+$ in the stated donor-only bulk.

Worked derivation: The charge density is

$$\rho = e(p + N_D^+ - n) \simeq 0.$$

Why: N-type means electron-majority conduction, not a macroscopic negative charge. Ionisation redistributes existing charge between mobile and lattice-associated forms.

Why not the others: (A) omits compensating donor charge. (B) confuses fixed ions with mobile electronic carriers. (C) reverses both donor-ion sign and majority-carrier type.

Key point: Majority carrier type and net bulk charge are distinct concepts.
